REVISED & EXPANDED EDITION • INCLUDES 28-DAY ACTION PLAN

DEEP SLEEP OPTIMIZATION

A SCIENCE-BACKED GUIDE TO RESTORATIVE REST AND WELLBEING



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HOW TO USE THIS GUIDE

Read it in an afternoon. Change your sleep in 28 days.

You can read this guide cover to cover in a few hours, but the real value comes from the 28 days that follow. Every chapter opens with a short summary, explains the science in plain language, and ends with one small, specific action you can take tonight or this week. Stack the changes gradually, in order, and many people notice a meaningful difference within two to three weeks.

Your 28-day roadmap

Week	Focus	What you will do	Daily time
1	**Light & timing**	Fix a wake time. Get outdoors within an hour of waking. Begin dimming lights two hours before bed.	~10 min
2	**Environment & routine**	Darken and cool the bedroom, move your phone out of bed, and build a 20–60 minute wind-down.	20–40 min
3	**Food, caffeine & movement**	Finish your main meal 3+ hours before bed, set a caffeine cut-off, and keep vigorous exercise 1+ hour away from bedtime.	No extra time
4	**Supplements & data**	Optionally trial one supplement. Review weekly trends, not nightly scores, and keep what works.	~5 min

START HERE

Before Week 1, spend three days building a baseline (it takes two minutes a day and is on the last page of Chapter 1). Without a starting point, you will not know what is working.

How to read the evidence

Sleep research ranges from large, replicated human trials to small or early studies. Where it helps, this edition marks how firm the evidence is. These are plain-language summaries, not formal grades.

●●● **Strong**	**Strong.** Consistent results in human studies and endorsed by clinical guidelines.
●●○ **Moderate**	**Moderate.** Promising and fairly consistent, but studies are smaller or results are mixed.
●○○ **Limited**	**Limited.** Early, small, short-term or mostly animal research. Low risk does not mean proven.

IMPORTANT

This guide is educational and does not replace medical care. If you suspect sleep apnoea, chronic insomnia, restless legs or another health condition, use it alongside professional support, not instead of it. If you are pregnant, take medication or have a medical condition, check with a doctor or pharmacist before trying supplements or light therapy.

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PART I

Understanding Deep Sleep

Why sleep quality matters more than hours, what deep sleep does, and what quietly erodes it.

CHAPTER 1

1 Why You're Exhausted — and Why It's Not Your Fault

You went to bed at a reasonable hour. You clocked something close to eight hours. Yet you woke up foggy, reached for coffee before you had brushed your teeth, and felt vaguely behind for the rest of the day. Sound familiar?

You are far from alone: surveys consistently find that many adults regularly wake unrefreshed even after enough hours in bed. For many, the core problem is not how long they sleep but how well: how regular, how deep and how undisturbed. The forces that erode it (artificial light, irregular schedules, late caffeine, alcohol, chronic stress) are woven into modern life, so this is not a willpower problem. Nothing here needs perfection or an expensive gadget, just curiosity and a few weeks of patience.

Three things worth knowing before you start

- **Hours are only the start.** Most adults need roughly 7–9 hours, but timing, regularity and continuity shape how restorative they are.
- **Small, unglamorous changes beat dramatic ones.** Light, a steady wake time and a cool, dark room beat most gadgets.
- **Give each change time.** Body clocks adjust over days to weeks. Judge progress on one- to two-week averages, not single nights.

> **IS IT REALLY YOUR SLEEP?**
>
> Persistent tiredness has many causes besides sleep habits, including sleep apnoea, anaemia, thyroid problems, low mood and some medications. If you still feel exhausted after three to four weeks of the basics here, speak to your doctor (see Chapter 10).

Your three-night baseline

Don't change anything yet. For three days, note the details below and rate how rested you feel a little while after waking.

Night	Lights out	Woke up	Last caffeine	Rested (1–10)
1				
2				
3				

CHAPTER 2

2 What Deep Sleep Actually Does

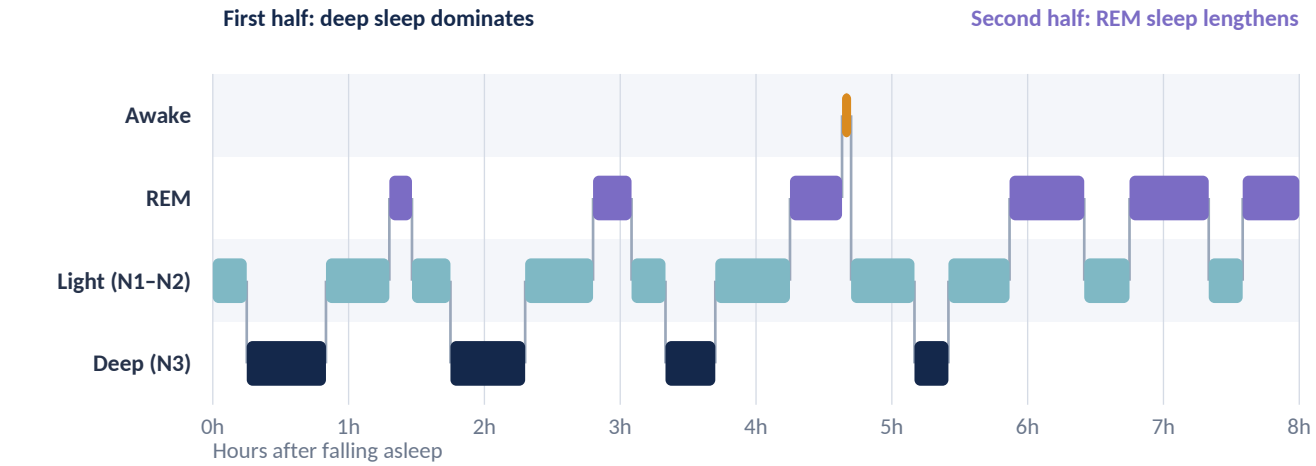
Sleep is not passive. While you lie unconscious, your brain and body run through an intricate sequence of stages, each doing work that waking life simply cannot.

THE SHORT VERSION

- Sleep runs in repeating cycles of about 90 minutes, moving between light, deep and REM sleep.
- Deep sleep dominates the first half of the night. REM sleep lengthens towards morning.
- You cannot force deep sleep, but you can create the conditions that let it happen.

The architecture of a night

Scientists divide sleep into two broad types. Non-REM (NREM) sleep progresses from light stages (N1–N2) to deep slow-wave sleep (N3). REM sleep, marked by rapid eye movements and brain activity that looks surprisingly like wakefulness, follows. A typical night cycles through these stages four to six times.



A stylised night for a healthy adult. Real nights vary, and brief awakenings are normal.

Stage	Common name	Typical share	What it does
N1	Light sleep	~2–5%	The doorway into sleep. Easily disturbed.
N2	Light sleep	~45–55%	Most of your night. Memory consolidation continues; the body cools and heart rate slows.
N3	Deep / slow-wave	~13–23%	Physical repair, immune function, growth hormone release and memory consolidation.
REM	Dream sleep	~20–25%	Emotional processing, learning and creative problem-solving; vivid dreams.

THE SCIENCE: DOES DEEP SLEEP REALLY "WASH" THE BRAIN?

In 2013, researchers reported that in sleeping mice the brain's glymphatic system (fluid-filled channels around blood vessels) cleared waste such as amyloid-beta faster than in awake mice (Xie et al.). A 2025 study in mice traced this to slow, rhythmic pulses of noradrenaline during non-REM sleep, and suggested that some sleeping pills may blunt them (Hauglund et al.).

The idea is still debated. A 2024 study found reduced clearance during sleep (Miao et al.), and the process has not been confirmed in humans. Treat brain "washing" as an exciting hypothesis, not a settled fact. The robust reasons to protect deep sleep are below.

How much deep sleep do you need?

Healthy adults typically spend about 13–23% of the night in deep sleep, roughly 1 to 1½ hours of a 7–8 hour night. The share falls with age, so people over 60 often get noticeably less, and genetics play a meaningful role too (Ohayon et al., 2004).

The key insight: chasing a specific deep-sleep number is less useful than creating the conditions that let your body reach deep sleep naturally and consistently. This guide is about those conditions.

What suffers when deep sleep is cut short

In a classic experiment, healthy volunteers whose slow-wave sleep was selectively suppressed for just three nights showed roughly a quarter lower insulin sensitivity, with no change in total sleep time (Tasali et al., 2008). More broadly, short or poor-quality sleep is linked to impaired glucose regulation, raised inflammation, a weaker immune response, and higher long-term risks of cardiovascular disease, type 2 diabetes and cognitive decline. Those long-term links are associations rather than proven causes, but small, consistent improvements compound over time.

THIS WEEK — TRY THIS

- [ ] If you use a tracker, look only at total sleep time and wake time. Ignore stage percentages for now.

CHAPTER 3

3 How Modern Life Quietly Undermines Sleep

Human sleep evolved in a world of firelight and dawn. We now live under LED screens, round-the-clock lighting, shift work and the expectation of being reachable at almost any hour. Each force tugs at a different thread of your sleep.

The sleep thief	How it hurts sleep	Where the fix lives
Light at the wrong times	Delays melatonin and shifts your body clock later	Chapters 4–5
Irregular schedules	Creates “social jet lag” and fragments the timing of deep sleep	Chapter 4
Caffeine	Masks sleep pressure and can lighten sleep for hours	Chapters 4 and 7
Alcohol	Helps you fall asleep, then fragments sleep and suppresses REM	Chapters 7 and 10
Chronic stress	Keeps the alert system switched on at night	Chapters 6 and 8

Light at the wrong times

Light is the main signal that synchronises your circadian clock to the 24-hour day. It does so partly by suppressing melatonin, a hormone that rises in darkness and starts the biological cascade towards sleep. Short-wavelength “blue” light, common in LED screens and overhead lighting, suppresses it most strongly.

> **THE SCIENCE: EVENING LIGHT AND MELATONIN**
>
> In a laboratory study, 12 adults read for four hours before bed on a light-emitting tablet, or from a printed book, across five evenings. Tablet reading suppressed melatonin by about 55%, delayed the body clock by roughly 1.5 hours, lengthened the time taken to fall asleep and left people less alert the next morning, even after eight hours in bed (Chang et al., 2015). Four hours is more than most people use, but the principle holds.

Irregular schedules

Your circadian rhythm does not know that Saturday is a day off. When your sleep and wake times on free days differ substantially from workdays, you impose a repeated time-zone shift on your body, which researchers call **social jet lag**. In a study of tens of thousands of people, each hour of social jet lag was associated with about a third higher odds of being overweight (Roenneberg et al., 2012), and later work links it to poorer mood and metabolic health.

Caffeine

Caffeine blocks adenosine receptors. Adenosine accumulates in the brain the longer you are awake, building *sleep pressure*. Caffeine does not clear adenosine; it just stops you feeling it. Its half-life averages five to six hours in adults (and varies a lot between individuals), so a 3 p.m. coffee can still have about half its caffeine circulating at 8–9 p.m.

THE SCIENCE: CAFFEINE CAN COST YOU MORE SLEEP THAN YOU NOTICE

In a placebo-controlled study, 400 mg of caffeine (about four small coffees) taken even six hours before bed reduced total sleep by more than an hour. Participants often did not notice the effect (Drake et al., 2013).

Alcohol

Alcohol is a sedative, so it can make it easier to fall asleep, and that is exactly why it is so counterproductive. As your body metabolises it, sleep becomes lighter and more fragmented, REM sleep is suppressed early in the night and early-morning waking becomes more likely (Ebrahim et al., 2013). The effect is dose-dependent and appears even at low intakes: in a study of about 4,000 adults wearing heart-rate-variability monitors, physiological recovery in the first hours of sleep fell by roughly 9 percentage points after low alcohol intake, 24 after moderate and 39 after high intake (Pietilä et al., 2018).

Chronic stress

Cortisol normally follows a daily curve: high in the morning, low at night. Prolonged stress can flatten that curve and keep the body's alert systems running at a low simmer. Racing thoughts at bedtime are both a symptom and a cause: they hold the nervous system in alert mode and stop the shift towards parasympathetic ("rest and digest") dominance that deep sleep needs.

THIS WEEK — TRY THIS

- [ ] Notice your screen and bright-light exposure in the two hours before bed.
- [ ] Note your last caffeine of the day, and the time.
- [ ] Still just observing. The changes start in the next chapter.

PART II

The Deep Sleep System

Four foundations — clock, environment, routine and fuel — in the order that works best.

CHAPTER 4

4 Set Your Clock: Light, Temperature & Timing

The four levers in this chapter are the foundation of everything else. Get these right and supplements, routines and gadgets become refinements rather than crutches.

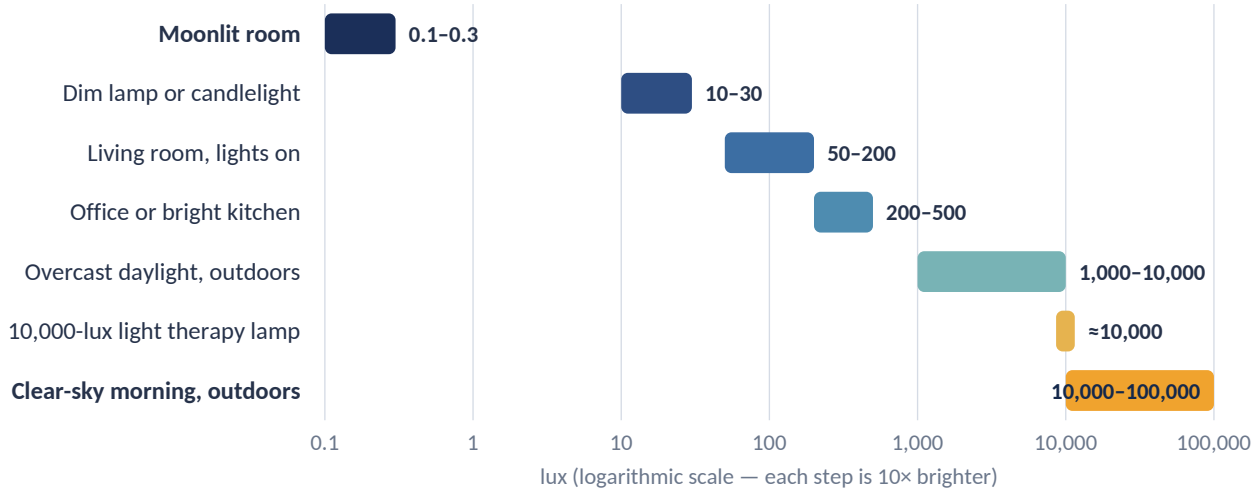
THE SHORT VERSION

- Bright light early, dim light late.
- Keep the bedroom cool, and use a warm bath or shower to speed the body's natural cooling.
- If you do only one thing from this guide: pick a fixed wake time and keep it within one hour, every day.

Lever 1 · Morning light: the master reset

Bright light soon after waking sets your circadian clock and determines when melatonin will rise that evening. It signals the suprachiasmatic nucleus (SCN), your brain's master clock, to start the day, and melatonin typically begins to rise about 13–15 hours later. The earlier and brighter the signal, the more reliably your sleep drive builds and your melatonin window opens on time.

Outdoor light on a clear morning reaches 10,000–100,000 lux. Indoor office lighting typically delivers 200–500 lux. That is why even a brief walk outside does more than sitting by a window.



Approximate light levels on a logarithmic scale. Even a grey day outdoors is many times brighter than most indoor lighting.

- [ ] Get outside within 30–60 minutes of waking.
- [ ] Aim for 5–10 minutes on bright days and 15–30 minutes when overcast.
- [ ] Skip sunglasses for this window (ordinary glasses and contact lenses are fine). Never look directly at the sun.
- [ ] If you cannot get outside, a 10,000-lux light therapy lamp for 20–30 minutes is an acceptable substitute. Check with a clinician first if you have an eye condition or bipolar disorder, or take photosensitising medication.

Lever 2 · Evening light: the "digital sunset"

The evening goal is to mimic what your ancestors experienced after sunset: a gradual fall in light intensity and a shift towards warmer light. That lets melatonin rise naturally.

Time before bed	Action
2 hours	Dim overhead lights. Switch to lamps below eye level.
1 hour	Move screens to night mode and cut brightness to around 50%. Keep them further from your face.
30 minutes	Ideally, screens off. Read something physical, stretch or talk.

THE SCIENCE: HOW DIM IS DIM, AND DO BLUE-LIGHT GLASSES HELP?

An international expert consensus recommends bright light by day (at least 250 "melanopic lux" at the eye), dim light in the three hours before bed (10 or less) and near-darkness during sleep (1 or less) (Brown et al., 2022). Most living rooms are brighter than that in the evening. The practical takeaway: lower, warmer and further from your eyes.

Blue-light-blocking glasses are not a shortcut. A 2023 Cochrane review found no clear benefit for sleep, though the evidence is low-certainty. Night mode helps a little, but brightness, distance and how stimulating the content is matter at least as much.

Lever 3 · Temperature

Falling asleep is partly triggered by a drop in core body temperature of about 1–2°C, so your bedroom should support that cooling.

- Target a room temperature of about 16–19°C (60–67°F) and experiment within that range. People differ, and those who run cold may prefer the warmer end.
- Use breathable natural-fibre bedding such as cotton, linen or bamboo.
- A warm bath or shower 1–2 hours before bed can help you fall asleep faster. Core temperature rises briefly, then drops as heat radiates from the skin (Haghayegh et al., 2019).

Lever 4 · The single most powerful schedule change

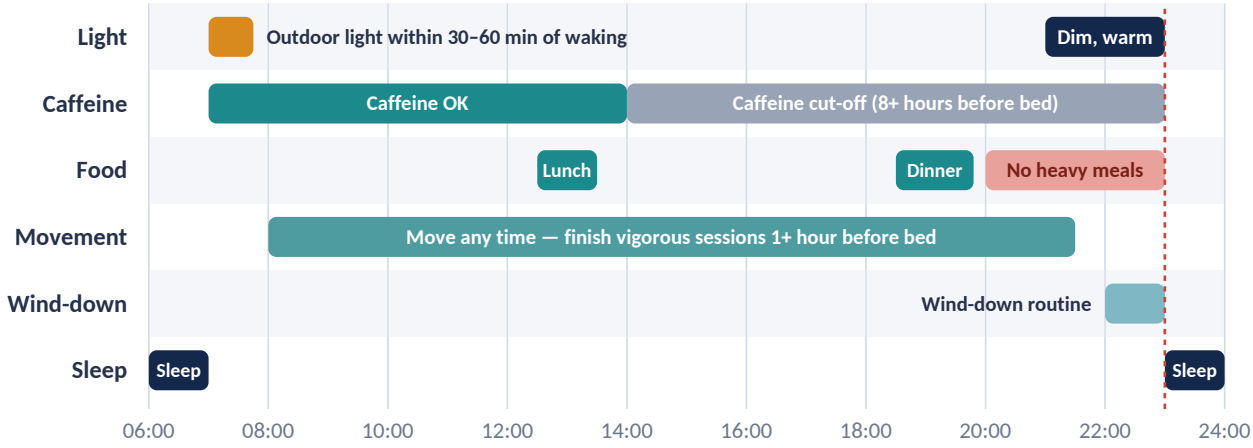
If you do only one thing from this guide, do this: pick a fixed wake-up time and keep it within one hour, seven days a week. Not because sleep gurus say so, but because your circadian rhythm is a biological clock, and it runs best with a consistent reference point. A stable wake time anchors the timing of your deepest sleep, steadies morning cortisol and melatonin timing, and builds sleep pressure more predictably, making it easier to fall asleep the following night.

- **Choosing your time:** pick the wake time your workdays already need, then protect it at weekends.

- **After a bad night:** get up at roughly your usual time, get morning light, and keep any nap under 30 minutes and before mid-afternoon.
- **Need to catch up?** Go to bed earlier rather than sleeping later.

Putting it together

A sample day · wake 07:00 · lights out 23:00 (shift everything to fit your own times)



A sample day. Everything shifts with your own wake time: count forward from waking and back from bedtime.

THIS WEEK — TRY THIS

- [ ] Choose a wake time you can hold even at weekends. Commit for 7 days.
- [ ] Get outside for morning light on at least 5 of those 7 days.
- [ ] Start dimming your home 2 hours before bed.

CHAPTER 5

5 Design Your Sleep Environment

Your bedroom should send one message to your nervous system: this is where we rest. Not where we work, scroll or process the day's stress. The most effective changes are often the cheapest.

THE SHORT VERSION

- Fix the environment in this order: dark, then cool, then quiet, then the "bed rule".
- Darkness is the highest-leverage place to start.

Darkness

Even dim light during sleep can suppress melatonin and activate alerting brain circuits. In a laboratory study, one night of sleeping in moderate room light (about 100 lux) raised heart rate during sleep and increased insulin resistance the next morning in healthy young adults (Mason et al., 2022).

- [ ] Use blackout curtains or a well-fitted eye mask.
- [ ] Cover or tape over LED indicator lights on chargers, routers and TVs.
- [ ] If you keep a clock in the room, turn it away from you.

Sound

Noise disrupts sleep even when it does not fully wake you. The brain keeps processing sound during N2 and REM sleep, and sudden sounds cause micro-arousals that fragment sleep without reaching consciousness.

- Rugs absorb sound, heavy curtains dampen street noise and foam earplugs block most environmental noise.
- Steady background sound (a fan, a white-noise machine or a free app) can mask irregular spikes. Evidence that white or pink noise improves sleep is mixed, so treat it as a masking tool and keep the volume low.

The bed rule

Stimulus control is one of the best-evidenced behavioural interventions for sleep. If you regularly do alerting or stressful things in bed (working, scrolling news, watching emotionally intense content) your brain learns to associate the bed with wakefulness. Use your bed only for sleep and intimacy. It sounds almost too simple. It works.

If your situation isn't ideal

Situation	What helps
Shared bed, different temperatures	Separate duvets or blankets so each person controls their own temperature.
Noisy street or neighbours	Heavy curtains, rugs, foam earplugs and steady masking sound.
Renting, or can't change the windows	Stick-on or tension-rod blackout blinds, or a comfortable eye mask.
One-room flat or studio	Use a lamp or fabric cover to make the bed area visually distinct, and avoid working on the bed.
Shift work	Treat your sleep block as fixed, make the room very dark, and limit bright light on the journey home.

A full printable checklist is in **Appendix B**.

THIS WEEK — TRY THIS

- [ ] Make one environment change tonight (darkness is the highest-leverage start).
- [ ] Plan one more change for later this week.

CHAPTER 6

6 Wind-Down Rituals That Actually Work

You cannot sprint to the end of the runway and expect to glide smoothly to a stop. The nervous system needs a genuine transition from “engaged” to “at rest”, and that transition takes time.

THE SHORT VERSION

- Pick a routine that fits the time you have tonight (20, 40 or 60 minutes) and keep the order the same.
- Repetition turns the routine into a cue: your brain learns what comes next.

What is actually happening

During wind-down you are trying to shift autonomic balance from sympathetic dominance (higher heart rate, cortisol and alerting noradrenaline) towards parasympathetic dominance (lower heart rate, falling core temperature and rising melatonin). Activities that are mentally engaging, emotionally stimulating or screen-based delay that shift, even when the content seems nominally relaxing.

Tools that work

Physical reading in low light
Fiction or non-urgent non-fiction. It avoids the reactive loop of news and social media, with no screen light and no notification pull.

Gentle movement or stretching
Light yoga or stretching engages slow, controlled breathing, activates the parasympathetic system and eases residual muscle tension.

Body scan or NSDR (limited–moderate evidence)
Non-sleep deep rest practices direct attention to physical sensations instead of thoughts. Mindfulness-based approaches have moderate evidence for improving sleep quality (Black et al., 2015). Body scans and yoga nidra are low-risk, but larger trials are needed.

Brain dump (moderate evidence)
Spend five minutes writing tomorrow’s to-do list. In one study, people who did this fell asleep about nine minutes faster than those who wrote about what they had already done (Scullin et al., 2018).

Three practical templates

20 min
The busy night

2 min	Dim the overhead lights
10 min	Light stretching or 4-7-8 breathing
5 min	Write tomorrow’s top 3 priorities
3 min	Brief body scan

40 min
The standard

10 min	Close work apps and tidy small things
15 min	Read something light
15 min	Guided NSDR or body-scan audio

60 min
The full reset

15 min	Gentle walk or movement
15 min	Warm shower or bath
15 min	Journalling or gratitude writing
15 min	NSDR or relaxation audio

Three techniques, step by step

4-7-8 breathing
Breathe in through the nose for 4 counts, hold for 7, breathe out slowly through the mouth for 8. Repeat 4 times. A long exhale is the active ingredient, so shorten the counts if it feels strained.

The physiological sigh
Take a full breath in through the nose, add a second short sip of air on top, then let out a long, slow breath through the mouth. Repeat 3–5 times. In a 2023 trial, five minutes a day of “cyclic sighing” improved mood and lowered resting breathing rate (Balban et al.).

Two-minute body scan
Lie down and move your attention slowly from your toes to your head, noticing sensations without trying to change them. When your mind wanders, gently return to the last spot.

THIS WEEK — TRY THIS

- [ ] Choose one template and use it on at least four nights this week.
- [ ] Adjust the timing to fit your life, but keep the basic shape and order.

CHAPTER 7

7 Food, Protein & Supplements

Nutrition's impact on sleep is real but secondary. It works best once the foundations of light, timing and environment are in place. Think of this chapter as fine-tuning, not a shortcut.

THE SHORT VERSION

- Finish your main meal about 3 hours before bed, and set a caffeine cut-off 8+ hours before bed.
- Pre-sleep protein helps muscle recovery. It is not a sleep aid.
- Supplements are optional, modest in effect and best trialled one at a time.

Meal timing

Digestion is metabolically demanding. A large meal close to bedtime can raise core body temperature (the opposite of what sleep needs), increase the chance of acid reflux and cause blood-glucose swings that fragment sleep later in the night. Finishing your main meal about three hours before bed is a widely used rule of thumb. The evidence is moderate, and individual responses vary. If you are genuinely hungry at bedtime, a small light snack beats lying awake hungry.

Your caffeine cut-off

Because caffeine lingers (Chapter 3), a good starting point is to stop at least 8 hours before bed, or about 2 p.m. for a 10–11 p.m. bedtime, and earlier if you are sensitive. Taper gradually rather than stopping abruptly to avoid withdrawal headaches.

Drink or food (typical serving)	Approx. caffeine
Brewed coffee, mug (240 ml)	~95 mg
Single espresso	~65 mg
Black tea (240 ml)	~45 mg
Cola (330 ml can)	~35 mg
Energy drink (250 ml)	~80 mg
Dark chocolate (30 g)	~20 mg

Amounts vary by brand and preparation. Decaffeinated coffee is not caffeine-free. NHS advice for pregnancy is to stay under 200 mg a day.

Movement

Regular physical activity is one of the most reliable ways to improve sleep, and a morning or afternoon walk outside doubles as light exposure. Moderate evening exercise is usually fine. Very vigorous exercise within an hour of bedtime can delay sleep onset for some people (Stutz et al., 2019), so keep hard sessions earlier.

Pre-sleep protein

For people who strength train or want better overnight recovery, pre-sleep protein has a meaningful evidence base. In one study, 40 g of casein before bed after evening resistance exercise increased overnight muscle protein synthesis by about 22% in young men (Res et al., 2012). Later research suggests around 30 g can work. Note that this is about muscle recovery, not sleep quality: there is no strong evidence that protein itself improves sleep. Total daily protein matters more than the pre-bed dose, and anyone with kidney disease should check with a doctor first.

Type	Digestion	Best for
Casein	Slow (several hours); forms a gel in the stomach	Muscle recovery; a longer overnight amino-acid supply
Whey isolate	Fast (about 1–2 hours); high in leucine	Lactose sensitivity; a lighter pre-bed option
Plant blend (pea + rice)	Medium; complete amino-acid profile	Dairy-free and vegan diets

Supplements: a sensible approach

Supplements will not fix an irregular schedule, evening light or late caffeine. If you want to try one, the options below are the most commonly used. Full doses, timing and cautions are in **Appendix D**.

Supplement	Typical dose	Evidence	Worth knowing
Magnesium (glycinate or threonate)	200–400 mg elemental	●○○ **Limited**	Small studies, mostly in older adults. Oxide is poorly absorbed.
L-theanine	100–200 mg	●○○ **Limited**	Calming without sedation; evidence for sleep itself is early.
Melatonin (low dose)	0.3–0.5 mg	●●○ **Moderate**	Best evidence is for shifting the body clock. A small effect on falling asleep.
Glycine	3 g	●○○ **Limited**	Small studies suggest better subjective sleep quality.

SUPPLEMENT SAFETY

Talk to a pharmacist or doctor first if you are pregnant or breastfeeding, take medication (especially sedatives, blood-pressure, diabetes or blood-thinning drugs), have kidney disease or another condition, or are considering a supplement for a child. Start with one at a time, trial it for at least 7 nights, and stop if you feel worse. Choose products with independent testing, because contents can differ from the label. Rules on melatonin vary by country: it is prescription-only in the UK and parts of Europe but sold as a supplement in the US.

THIS WEEK — TRY THIS

- ☐ Move your main meal to at least 3 hours before bed if it is currently later.
- ☐ Set a caffeine cut-off (a good starting point: no caffeine after 2 p.m.).
- ☐ If considering a supplement, start with one and trial it for at least 7 nights before judging.

PART III

Mind, Data & Long-Term Change

Quieting a busy mind, using trackers wisely, and knowing when to ask for help.

CHAPTER 8

8 Calm the “Wired but Tired” Mind

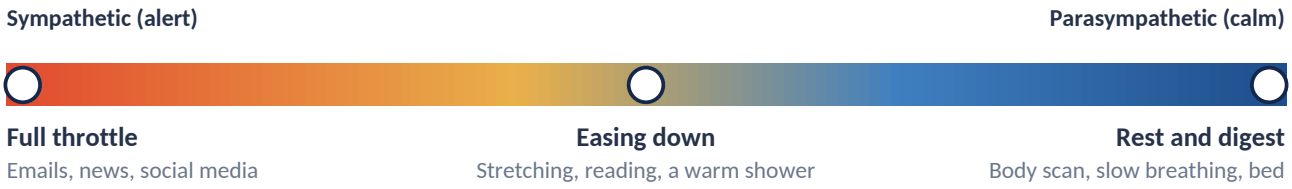
Perhaps the most frustrating sleep experience: you are exhausted, you have done everything right, you lie down, and your brain decides this is an excellent time to review every unresolved problem from the past six months.

THE SHORT VERSION

- Trying harder to sleep backfires. Aim to create the conditions, then let go.
- Stimulus control, scheduled worry time and defusion are the best-evidenced mind tools.
- Have a plan for the 3 a.m. wake-up so it does not become a crisis.

Why trying harder makes it worse

Sleep is paradoxical: the harder you try to force it, the more elusive it becomes. Researchers call the mental and emotional work of trying to make yourself sleep “sleep effort”, and it activates the arousal system, raising heart rate and cortisol, the opposite of the physiology needed for sleep onset (Espie et al., 2006). This is one reason people with chronic insomnia often show signs of hyperarousal, such as higher heart rate and metabolic rate across the day and night, despite feeling exhausted (Bonnet & Arand, 1995).



A wind-down moves you along this dial, from alert to calm.

Evidence-based mind tools

Stimulus control therapy

●●● **Strong**

One of the most replicated non-drug interventions for insomnia. By repeatedly pairing the bed with sleepiness instead of wakefulness, you rebuild the automatic association, often within a few weeks. The first nights can feel harder before they feel better.

1	**Use the bed only for sleep and intimacy.**
2	**Go to bed only when sleepy** (not just tired).
3	**If you have been awake for what feels like 20 minutes, get up** Go to another room and do something calm in dim light until sleepy. Avoid clock-watching.
4	**Return to bed when sleepy and repeat as needed.**
5	**Keep your wake time fixed** regardless of how the night went.

Scheduled worry time

●●○ **Moderate**

Set aside 15–20 minutes earlier in the evening (not in bed and not in the last hour before sleep) to write down current worries and the next small action for each. When a thought appears at bedtime you can genuinely tell yourself: *“I have a time for this. It is handled enough for tonight.”*

Cognitive defusion

●●○ **Moderate**

From Acceptance and Commitment Therapy: rather than fighting thoughts or trying to replace them, observe them as mental events without engaging their content. “There is the work worry again” rather than “I need to solve this right now.” It reduces emotional reactivity without requiring thought suppression, which tends to backfire.

For breathing exercises and body scans, see the step-by-step techniques in Chapter 6.

WHEN YOU WAKE AT 3 A.M.

1. Do not check the time. **2.** Try slow breathing with a long exhale, or a body scan. **3.** If you are still wide awake after about 20 minutes, get up, keep the lights low and do something calm without a phone until sleepy. **4.** Remind yourself that one rough night does not undo your progress. Keep your wake time.

THIS WEEK — TRY THIS

- [ ] Choose one tool from this chapter and practise it 3–5 nights, even on nights when sleep feels fine, so your brain learns the pattern.
- [ ] If you find yourself clock-watching, turn the clock face away.

CHAPTER 9

9 Using Trackers Without Obsessing

Wearables and sleep apps can be genuinely useful. They can also become a source of anxiety that makes your sleep worse. A 2017 paper in the *Journal of Clinical Sleep Medicine* coined the term **orthosomnia** for sleep anxiety that develops from the pursuit of better tracker scores (Baron et al., 2017).

THE SHORT VERSION

- Trust trackers for trends in timing and total sleep, not nightly grades or stage percentages.
- Review the data once a week. On other days, glance at total sleep and close the app.

What trackers do well

• Revealing trends in total sleep time over weeks.
• Showing how consistent your bedtime and wake time really are, compared with how consistent you think they are.
• Highlighting correlations, such as late alcohol reliably raising your resting heart rate that night.

What trackers do poorly

• Measuring sleep stages. Wearables infer them from movement and heart-rate signals, not the brain waves (EEG) used in sleep labs.
• Detecting quiet wakefulness. They tend to overestimate sleep.
• Being consistent with each other. Illustratively, a device might report 45 minutes of deep sleep when a lab records 80.

Validation studies consistently find that consumer devices are reasonable at estimating total sleep time but substantially less accurate at staging. Use stage data as a rough directional signal over weeks, not as a nightly diagnosis.

A saner approach to data

Metric	How to use it
Total sleep time	Track nightly. The weekly average is the most meaningful number.
Wake-time consistency	Your anchor. Protect this above all else.
Bedtime consistency	Aim for 30 minutes or less of variation. If wider, investigate why.
Resting heart rate / HRV (if available)	A useful proxy for recovery. Trends matter, not single nights.
Deep / REM percentages	Interesting directional data only. Do not optimise them directly.

Your five-minute weekly review

Question	My answer
What was my average sleep time this week?	
How much did my wake time vary?	
Which habit did I keep most consistently?	
What is one small change for next week?	

THIS WEEK — TRY THIS

- [ ] Designate one day a week (for example, Sunday morning) as your data review day.
- [ ] On all other days, check total sleep time only, then close the app.

CHAPTER 10

10 Troubleshooting, Plateaus & When to Get Help

Progress with sleep is rarely linear. You will have rough stretches: travel, illness, work pressure, a newborn, grief. The goal is not to never have a bad night. It is to have a system you can return to.

Quick troubleshooting

If this is happening...	Common causes	Try first
Taking more than 30 minutes to fall asleep	Late caffeine, bright evenings, going to bed before you are sleepy, worry	Check your caffeine cut-off and evening light. Use stimulus control: get up at about the 20-minute mark rather than lying awake.
Waking at 2–4 a.m. and unable to return to sleep	Alcohol (even 1–2 drinks), late eating, a warm room, stress, needing the toilet, pain or reflux, low mood	Review alcohol and meal timing, cool the room and use your 3 a.m. plan (Chapter 8). If it persists, mention it to your doctor.
Waking exhausted despite 8+ hours in bed	Sleep apnoea, which affects an estimated 1 billion people worldwide and is widely undiagnosed (Benjafield et al., 2019). Also depression, thyroid problems, anaemia, medication effects	Seek medical evaluation, especially if you snore, gasp or feel sleepy in the day.
Plateau after initial improvement	Light and timing habits slip gradually	Normal. Revisit the Week 1 foundations first, and only add new things once they are solid.

Still not sleeping well? Start here

Work down the list. Stop at the first "no" and fix that before moving on.

1	**Is your sleep schedule consistent?** Regular bedtime and wake time, including weekends. *If no: set a fixed wake time (Chapter 4).*
2	**Are caffeine and alcohol limited, especially late in the day?** *If no: set earlier cut-offs (Chapters 3 and 7).*
3	**Is your bedroom cool, dark and quiet?** *If no: work through the environment checklist (Chapter 5 and Appendix B).*
4	**Do you get daylight by day and dim light by night?** *If no: morning light and a digital sunset (Chapter 4).*
5	**Are stress or worry keeping you awake?** *If yes: wind-down, worry time and stimulus control (Chapters 6 and 8). Consider counselling.*
6	**Do you have symptoms of a medical cause, or insomnia lasting months?** Loud snoring, leg sensations, severe daytime sleepiness. *If yes: talk to a professional.*

WHEN TO SEEK PROFESSIONAL HELP

This guide is for lifestyle-based sleep difficulties. Some conditions need professional diagnosis and treatment that no habit change will replace. Speak to a health professional if:

- You snore loudly, wake gasping, or have been told you stop breathing during sleep.
- You feel dangerously sleepy while driving or in safety-critical situations. Do not drive while drowsy.
- Restless or uncomfortable sensations in your legs disrupt sleep (possible restless legs syndrome).
- You have insomnia at least 3 nights a week for more than 3 months despite consistent effort.
- You rely on alcohol or sleeping tablets to get to sleep.
- You feel persistently low, hopeless or anxious alongside poor sleep. These may be a cause rather than a consequence, and are treatable.

CBT-I: THE FIRST-LINE TREATMENT

Cognitive behavioural therapy for insomnia (CBT-I) is recommended as the first-line treatment for chronic insomnia by the American College of Physicians (Qaseem et al., 2016) and other bodies. In the long term it works better than sleeping tablets, with fewer side effects, though some techniques can cause temporary daytime sleepiness at first. It is available with a therapist, in groups, or through validated digital programmes such as Sleepio or the free CBT-i Coach app. Your GP can advise on local options.

THIS WEEK — TRY THIS

- [ ] Note any recurring issues that are not resolving after 3–4 weeks of consistent effort.
- [ ] If anything from the list above applies to you, book that appointment.

PUTTING IT ALL TOGETHER

Your Sleep System on One Page

Everything in this guide comes back to five foundations. If life gets busy or progress stalls, return to this list.

1	**A fixed wake time** Within one hour, seven days a week. This anchors everything else.
2	**Morning light** Outdoors within 30–60 minutes of waking, for 5–30 minutes depending on brightness.
3	**A caffeine cut-off** At least 8 hours before bed (about 2 p.m. for most people), and little or no alcohol close to bedtime.
4	**A dim, dark, cool bedroom** Dim the home two hours before bed, then sleep in a cool, dark, quiet room.
5	**A repeatable wind-down** 20, 40 or 60 minutes, the same order each night, with a plan for 3 a.m. wake-ups.

A realistic weekly rhythm

When	What
Every day	Same wake time. Morning light. Caffeine cut-off. Wind-down. Dim lights in the evening.
Once a week	Five-minute data review. Pick one small change for the week ahead.
When sleep slips	Go back to Week 1: wake time and light. Add things back one at a time.
If nothing helps	Speak to your doctor, and ask about CBT-I.

A FINAL THOUGHT

You do not need a perfect night, a perfect score or a perfect routine. You need a system you can return to, a few weeks of patience, and a little curiosity about what works for your body. Start tonight with one change.

TOOLS

Appendices

Printable plans, checklists and quick-reference tables.

APPENDIX A

A 28-Day Deep Sleep Action Plan

Keep earlier habits going as you add new ones. Tick the boxes as you go; the plan builds week by week.

WEEK 1 Light & Timing — anchor your clock

Goal: A stable wake time and strong morning light signal.

Day 1	Choose your fixed wake time and set the alarm. Commit to it for the rest of the plan.
Days 1–7	Hold your wake time within 1 hour. Get outside within 60 minutes of waking on at least 5 days.
Days 4–7	Begin dimming lights and switching screens to night mode 2 hours before bed.

Daily check	D1	D2	D3	D4	D5	D6	D7
Woke within 1 hour of target	☐	☐	☐	☐	☐	☐	☐
Outdoor morning light	☐	☐	☐	☐	☐	☐	☐
Dimmed lights 2 hours before bed	☐	☐	☐	☐	☐	☐	☐

WEEK 2 Environment & Routine — optimise your sleep space

Goal: A dark, cool bedroom and a repeatable wind-down, on top of Week 1 habits.

Day 8	Add a blackout solution (curtains, blinds or eye mask).
Day 9	Cover or remove LED indicators and move your phone out of bed.
Days 10–14	Use one wind-down routine (20, 40 or 60 minutes) on at least 4 nights.

Daily check	D8	D9	D10	D11	D12	D13	D14
Wake time and morning light	☐	☐	☐	☐	☐	☐	☐
Dark, cool room; phone out of bed	☐	☐	☐	☐	☐	☐	☐
Wind-down routine done	☐	☐	☐	☐	☐	☐	☐

WEEK 3 Food, Caffeine & Movement — adjust timing

Goal: Meal and caffeine timing that protect deep sleep, while keeping earlier habits.

Day 15	Move your main meal to at least 3 hours before bed.
Day 16	Set a caffeine cut-off (start with 2 p.m.).
Days 17–21	Observe energy, mood and sleep. Adjust the cut-off by an hour if needed, and keep vigorous exercise 1+ hour before bed.

Daily check	D15	D16	D17	D18	D19	D20	D21
Main meal 3+ hours before bed	☐	☐	☐	☐	☐	☐	☐
Caffeine before cut-off	☐	☐	☐	☐	☐	☐	☐
Wind-down routine done	☐	☐	☐	☐	☐	☐	☐
Rested (1–10)							

WEEK 4 Supplements & Data — fine-tune and review

Goal: Decide what to keep. Supplements are optional.

Day 22	Decide whether you want a supplement. If so, choose one (magnesium glycinate is a common starting point, although evidence is limited) and start tonight.
Days 22–28	Trial it for 7 nights. Skip this step entirely if you prefer. Keep tracking your foundations.
Day 28	Do your full weekly data review. Identify the 1–2 changes with the clearest benefit and make them permanent.

Daily check	D22	D23	D24	D25	D26	D27	D28
Foundations (wake time, light, wind-down)	☐	☐	☐	☐	☐	☐	☐
Supplement taken (if using)	☐	☐	☐	☐	☐	☐	☐
Rested (1–10)							

APPENDIX B

B Sleep Environment Checklist

Tick what is already in place and pick one thing to change this week. Darkness first.

DARK

- [ ] Blackout curtains, blinds or a well-fitted eye mask
- [ ] LED indicator lights covered or removed
- [ ] Clock turned away, or out of sight
- [ ] Night lights (if needed) are dim, warm and low

COOL

- [ ] Room feels cool (16–19°C / 60–67°F)
- [ ] Breathable natural-fibre bedding
- [ ] Separate blankets if partners prefer different temperatures

QUIET

- [ ] Noise is low or masked by steady background sound
- [ ] Heavy curtains or rugs soften street noise
- [ ] Earplugs available for noisy nights

COMFORT & CUES

- [ ] Mattress and pillow support comfortable spinal alignment
- [ ] No screens used in bed for work or social media
- [ ] Phone on silent or do-not-disturb, or in another room
- [ ] Bed used only for sleep and intimacy

My score: _____ / 14 **My first change:** _____

Changes I made and what I noticed

Date	Change	What I noticed

APPENDIX C

C Wind-Down Routine Builder

Pick two or three activities from the menu, fit them to tonight's time, and write your routine below. Keep the same order each night.

Lower the lights	Move	Unload	Settle
Dim overhead lights Switch to lamps Screens to night mode	Gentle walk Light stretching or yoga Warm shower or bath	Tomorrow's top 3 priorities Worry list (earlier in the evening) Tidy one small thing	Read a physical book 4-7-8 or sigh breathing Body scan or NSDR audio

My routine

Step	Activity	Minutes
1		
2		
3		
4		
5		

Start time: _____ **Lights out:** _____ **Total:** _____ minutes

QUICK REMINDER

20 minutes — dim lights, stretch or 4-7-8 breathing, top 3 priorities, body scan. **40 minutes** — tidy up, read, guided NSDR. **60 minutes** — walk, warm shower, journalling, NSDR. See Chapter 6 for details.

APPENDIX D

D Supplement Quick Reference

Supplements are optional and modest in effect. Evidence is rated as in “How to read the evidence”. Trial one at a time for at least 7 nights.

Supplement	How it may help	Dose and timing	Evidence	Cautions
Magnesium glycinate or threonate	May support relaxation and the stress response. Small studies, mostly in older adults with insomnia.	200–400 mg elemental magnesium, 1–2 hours before bed	●○○ **Limited**	Can cause loose stools. Oxide is poorly absorbed. Kidney disease and some antibiotics or osteoporosis drugs interact. Keep supplemental intake to roughly 350–400 mg a day.
L-theanine	Calming without sedation. Evidence for sleep itself is early.	100–200 mg, 30–60 minutes before bed	●○○ **Limited**	Generally well tolerated. May lower blood pressure slightly.
Melatonin low dose	Signals darkness to the body clock. Best evidence is for jet lag and shifting timing; a small effect on falling asleep.	0.3–0.5 mg, 30–60 minutes before target bedtime. Timing for shifting the clock is more specific, so ask a clinician.	●●○ **Moderate**	Possible drowsiness or vivid dreams. Avoid in pregnancy or breastfeeding, and with some conditions or medicines, unless advised. Not for children without medical advice. Label doses can be inaccurate.
Glycine	May lower core temperature and improve subjective sleep in small studies.	3 g before bed	●○○ **Limited**	Generally well tolerated.

APPROACH WITH CAUTION

Over-the-counter antihistamine sleep aids can cause next-day drowsiness and are not recommended for regular use. **Alcohol** disrupts sleep (Chapter 3). **Sleeping tablets** should only be used as advised by a doctor, and generally short-term. Herbal and cannabis-related products vary widely in dose and quality, and evidence is limited.

APPENDIX E

E References & Further Reading

The sources below underpin the claims in this guide. Several statements were refined or qualified in this edition to match what the studies actually found.

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